Started on	Tuesday, 29 November 2022, 5:15 PM
State	Finished
Completed on	Tuesday, 29 November 2022, 5:17 PM
Time taken	2 mins 7 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 2.00 out of 2.00

Which of the following synchronization methods uses busy waiting?

Select one:

- a. spinlock ✔
- b. mutex
- c. semaphore
- d. condition variables

The `lock()` operation on a spinlock triggers a busy waiting if the lock is being held by another thread.

The correct answer is: spinlock

Question **2**

Correct

Mark 2.00 out of 2.00

Which of the following synchronization methods uses `acquire()` and `release()`?

Select one:

- a. mutex
- b. semaphore ✔
- c. condition variable
- d. atomic variables

A semaphore uses the `acquire()` / `release()` calls.

The correct answer is: semaphore

Question **3**

Correct

Mark 2.00 out of 2.00

What type of race condition can happen when running the below code snippet in multiple threads?

```
1 if (var < n)
2     var++;
```

Select one:

- a. time of check to time of use ✔
- b. data leak
- c. use after free
- d. dangling pointer

TOCTOU is a type of race condition which involves checking a shared (unprotected) variable value after it has been modified.

The correct answer is: time of check to time of use

Question **4**

Correct

Mark 2.00 out of 2.00

Which of the following refers to copy-on-write (COW)?

Select one:

- a. When we reach the limit of physical memory and we swap out a memory page to the disk.
- b. When we try to reach an invalid page and a segmentation fault occurs.
- c. When the Kernel duplicates the contents of a physical page (frame) as a result of writing in one of the child's pages. ✔
- d. When we try to reach a page which has been swapped out and it gets swapped back in

Copy-on-write (COW) means the address space of the parent and the child point to the same physical memory initially (i.e. after `fork()` returns).
The physical pages (frames) are only duplicated when a write is performed in the child.

The correct answer is: When the Kernel duplicates the contents of a physical page (frame) as a result of writing in one of the child's pages.

Question **5**

Correct

Mark 2.00 out of 2.00

What will the following code snippet print when `fork()` succeeds?

```
int i = 0;
if (fork() == 0)
    i++;
else
    sleep(1);
printf("%d\n",i);
```

Select one:

- a. It will print 0 (from the child process) followed by 0 (from the parent).
- b. It will print 0 (from the child process) followed by 1 (from the parent).
- c. It will print 1 (from the child process) followed by 0 (from the parent). ✔
- d. It will print 1 (from the child process) followed by 1 (from the parent).

If the `fork()` call will be successful, the value returned by the function will be 0 in the child process and non zero in the parent process. As a consequence, the child process will increment the variable and the parent will sleep for 1 second. We can assume that the child process will print 1 while the parent is asleep. After waking up, the parent will print its own value of i which is not shared with the child process and therefore will be 0 because it wasn't incremented

The correct answer is: It will print 1 (from the child process) followed by 0 (from the parent).

◄ Curs 06: Calcul: Recap

Jump to...

Curs 08: I/O (partea 1) ►

Started on	Tuesday, 29 November 2022, 5:16 PM
State	Finished
Completed on	Tuesday, 29 November 2022, 5:18 PM
Time taken	1 min 28 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 2.00 out of 2.00

How can we have a more cpu-intensive system?

Select one:

- a. Increase the priority of the processes in READY state.
- b. Decrease the time slice of the processes in RUNNING state.
- c. Decrease the priority of CPU-intensive processes.
- d. Increase the time slice of processes. ✔

Increasing the time slice of processes will reduce the number of context switches and thus lower the overhead and increase the productivity of CPU-intensive systems.

The correct answer is: Increase the time slice of processes.

Question **2**

Correct

Mark 2.00 out of 2.00

Which of the following refers to copy-on-write (COW)?

Select one:

- a. When we reach the limit of physical memory and we swap out a memory page to the disk.
- b. When we try to reach an invalid page and a segmentation fault occurs.
- c. When the Kernel duplicates the contents of a physical page (frame) as a result of writing in one of the child's pages. ✔
- d. When we try to reach a page which has been swapped out and it gets swapped back in

Copy-on-write (COW) means the address space of the parent and the child point to the same physical memory initially (i.e. after `fork()` returns).
The physical pages (frames) are only duplicated when a write is performed in the child.

The correct answer is: When the Kernel duplicates the contents of a physical page (frame) as a result of writing in one of the child's pages.

Question **3**

Correct

Mark 2.00 out of 2.00

Which of the following triggers a context switch:

Select one:

- a. A RUNNING process creates a new thread
- b. A process new process is created with a higher priority than the one in READY state
- c. A process moves from the WAITING state to READY state
- d. A RUNNING process performs a blocking IO operation ✔

A RUNNING process performing a blocking I/O operation will switch its state from RUNNING to WAITING(BLOCKED) causing a context switch.

The correct answer is: A RUNNING process performs a blocking IO operation

Question **4**

Correct

Mark 2.00 out of 2.00

What memory zone is not shared between threads of the same process?

Select one:

- a. stack ✔
- b. heap
- c. data
- d. rodata

Each thread has its own stack zone.

The correct answer is: stack

Question **5**

Correct

Mark 2.00 out of 2.00

Which of the following situations would cause a deadlock?

Select one:

- a. Two threads try to modify the same variable at the same time
- b. A condition variable is never met
- c. A process waits at a barrier
- d. Two threads wait for each other to release locks ✔

Two threads waiting for each other to release locks cause a deadlock.

The correct answer is: Two threads wait for each other to release locks

◄ Curs 06: Calcul: Recap

Jump to...

Curs 08: I/O (partea 1) ►

Started on	Tuesday, 29 November 2022, 5:22 PM
State	Finished
Completed on	Tuesday, 29 November 2022, 5:24 PM
Time taken	2 mins 3 secs
Grade	**9.00** out of 10.00 (**90**%)

Question **1**

Correct

Mark 2.00 out of 2.00

Which of the following synchronization methods uses `lock()` and `unlock()`?

Select one:

- a. mutex ✔
- b. semaphore
- c. barrier
- d. atomic variables

A mutex uses the lock / unlock calls.

The correct answer is: mutex

Question **2**

Partially correct

Mark 1.00 out of 2.00

What is the key difference between locks and notifications?

Select one or more:

- a. Unlock must be called by the thread who holds the lock while notifications can be sent from any thread ✔
- b. Notifications are prone to race conditions
- c. Notify must be called by the thread who holds the condition/semaphore lock while unlocks can be sent from any thread

Unlock must be called by the thread who holds the lock while notifications can be sent from any thread

The correct answers are: Unlock must be called by the thread who holds the lock while notifications can be sent from any thread, Notifications are prone to race conditions

Question **3**

Correct

Mark 2.00 out of 2.00

How does the child process(created through `fork()`) differ from its parent?

Select one:

- a. file descriptors
- b. process id ✔
- c. code area
- d. user which created the process

The `fork()` call creates a new process which is an identical copy of the parent process and inherits file descriptors, the memory of the process (including the code) and the user which created the process. It does not inherit the process id.

The correct answer is: process id

Question **4**

Correct

Mark 2.00 out of 2.00

Which of the following triggers a context switch:

Select one:

- a. A RUNNING process creates a new thread
- b. A process new process is created with a higher priority than the one in READY state
- c. A process moves from the WAITING state to READY state
- d. A RUNNING process performs a blocking IO operation ✔

A RUNNING process performing a blocking I/O operation will switch its state from RUNNING to WAITING(BLOCKED) causing a context switch.

The correct answer is: A RUNNING process performs a blocking IO operation

Question **5**

Correct

Mark 2.00 out of 2.00

What will the following code snippet print when `fork()` succeeds?

```
int i = 0;
if (fork() == 0)
    i++;
else
    sleep(1);
printf("%d\n",i);
```

Select one:

- a. It will print 0 (from the child process) followed by 0 (from the parent).
- b. It will print 0 (from the child process) followed by 1 (from the parent).
- c. It will print 1 (from the child process) followed by 0 (from the parent). ✔
- d. It will print 1 (from the child process) followed by 1 (from the parent).

If the `fork()` call will be successful, the value returned by the function will be 0 in the child process and non zero in the parent process. As a consequence, the child process will increment the variable and the parent will sleep for 1 second. We can assume that the child process will print 1 while the parent is asleep. After waking up, the parent will print its own value of i which is not shared with the child process and therefore will be 0 because it wasn't incremented

The correct answer is: It will print 1 (from the child process) followed by 0 (from the parent).

◄ Curs 06: Calcul: Recap

Jump to...

Curs 08: I/O (partea 1) ►

Started on	Tuesday, 29 November 2022, 5:11 PM
State	Finished
Completed on	Tuesday, 29 November 2022, 5:14 PM
Time taken	2 mins 39 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 2.00 out of 2.00

Which of the following synchronization methods uses busy waiting?

Select one:

- a. spinlock ✔
- b. mutex
- c. semaphore
- d. condition variables

The `lock()` operation on a spinlock triggers a busy waiting if the lock is being held by another thread.

The correct answer is: spinlock

Question **2**

Correct

Mark 2.00 out of 2.00

Which of the following synchronization methods uses `acquire()` and `release()`?

Select one:

- a. mutex
- b. semaphore ✔
- c. condition variable
- d. atomic variables

A semaphore uses the `acquire()` / `release()` calls.

The correct answer is: semaphore

Question **3**

Correct

Mark 2.00 out of 2.00

Which of the following situations would cause a deadlock?

Select one:

- a. Two threads try to modify the same variable at the same time
- b. A condition variable is never met
- c. A process waits at a barrier
- d. Two threads wait for each other to release locks ✔

Two threads waiting for each other to release locks cause a deadlock.

The correct answer is: Two threads wait for each other to release locks

Question **4**

Correct

Mark 2.00 out of 2.00

Which of the following are disadvantages of a spinlock when compared to a mutex?

Select one:

- a. Spinlocks are harder to implement than mutexes.
- b. Spinlocks are always slower than mutexes.
- c. Spinocks occupy CPU time doing nothing. ✔
- d. Spinlocks can create racing conditions.

Spinlocks perform busy waiting which involves running no-ops and thus occupying CPU load by doing nothing.

The correct answer is: Spinocks occupy CPU time doing nothing.

Question **5**

Correct

Mark 2.00 out of 2.00

On an N core system, how many processes can we have in the READY state?

Select one:

- a. 1
- b. 0
- c. N
- d. any ✔

We can have any number of processes in the READY state, we only have a limit on the number of RUNNING processes.

The correct answer is: any

◄ Curs 06: Calcul: Recap

Jump to...

Curs 08: I/O (partea 1) ►

Started on	Tuesday, 29 November 2022, 5:19 PM
State	Finished
Completed on	Tuesday, 29 November 2022, 5:21 PM
Time taken	2 mins 38 secs
Grade	**9.00** out of 10.00 (**90**%)

Question **1**

Correct

Mark 2.00 out of 2.00

Which of the following refers to copy-on-write (COW)?

Select one:

- a. When we reach the limit of physical memory and we swap out a memory page to the disk.
- b. When we try to reach an invalid page and a segmentation fault occurs.
- c. When the Kernel duplicates the contents of a physical page (frame) as a result of writing in one of the child's pages. ✔
- d. When we try to reach a page which has been swapped out and it gets swapped back in

Copy-on-write (COW) means the address space of the parent and the child point to the same physical memory initially (i.e. after `fork()` returns).
The physical pages (frames) are only duplicated when a write is performed in the child.

The correct answer is: When the Kernel duplicates the contents of a physical page (frame) as a result of writing in one of the child's pages.

Question **2**

Correct

Mark 2.00 out of 2.00

What memory zone is not shared between threads of the same process?

Select one:

- a. stack ✔
- b. heap
- c. data
- d. rodata

Each thread has its own stack zone.

The correct answer is: stack

Question **3**

Correct

Mark 2.00 out of 2.00

How does the child process(created through `fork()`) differ from its parent?

Select one:

- a. file descriptors
- b. process id ✔
- c. code area
- d. user which created the process

The `fork()` call creates a new process which is an identical copy of the parent process and inherits file descriptors, the memory of the process (including the code) and the user which created the process. It does not inherit the process id.

The correct answer is: process id

Question **4**

Correct

Mark 2.00 out of 2.00

What type of race condition can happen when running the below code snippet in multiple threads?

```
1 if (var < n)
2     var++;
```

Select one:

- a. time of check to time of use ✔
- b. data leak
- c. use after free
- d. dangling pointer

TOCTOU is a type of race condition which involves checking a shared (unprotected) variable value after it has been modified.

The correct answer is: time of check to time of use

Question **5**

Partially correct

Mark 1.00 out of 2.00

What is the key difference between locks and notifications?

Select one or more:

- [x] a. Unlock must be called by the thread who holds the lock while notifications can be sent from any thread ✔
- [ ] b. Notifications are prone to race conditions
- [ ] c. Notify must be called by the thread who holds the condition/semaphore lock while unlocks can be sent from any thread

Unlock must be called by the thread who holds the lock while notifications can be sent from any thread

The correct answers are: Unlock must be called by the thread who holds the lock while notifications can be sent from any thread, Notifications are prone to race conditions

◄ Curs 06: Calcul: Recap

Jump to...

Curs 08: I/O (partea 1) ►